Project Synopsis — Submission Template

HCLTech Career Shaper Programme | III Year Industry Project | Academic Year 2025–26

Project Title	Crop Disease Detection Using Convolutional Neural Networks
Project No.	35
Domain	Digital-Data-AI / AIML
Difficulty & Duration	Advanced | 4 Weeks
Format	Individual Project
Learner Name	**Monika Yadav**
Roll Number	**2400320109014**
College / Campus	**ABES Engineering College**
Mentor Name	**kathiresan**
Submission Date	**28 October 2026**

1. Problem Statement (5W1H Framework)

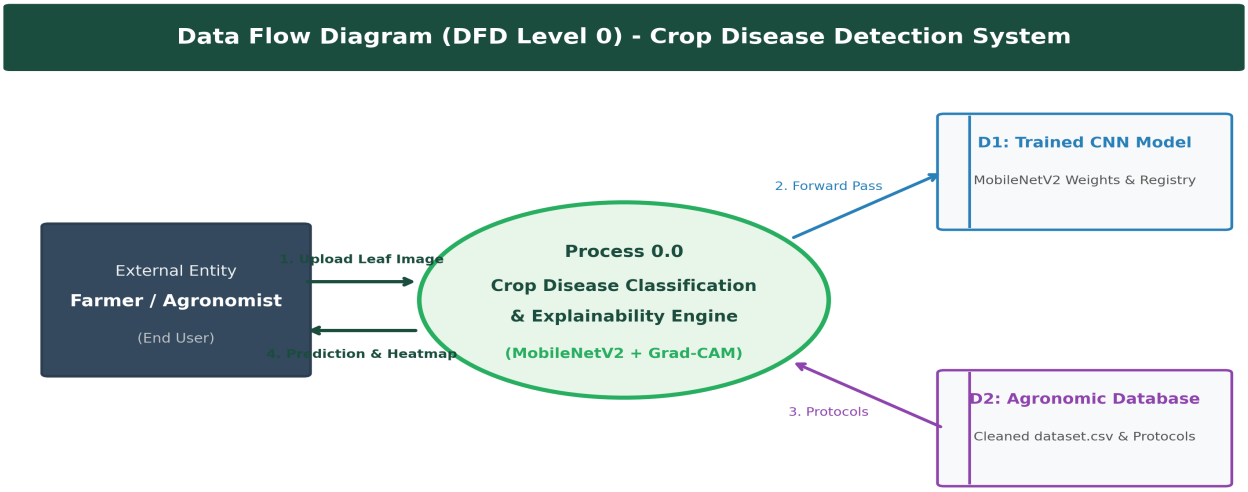
Smallholder farmers, greenhouse operators, and agricultural extension scouts (**Who**) face devastating crop failures caused by foliar fungal and bacterial plant pathogens that are identified either too late or inaccurately (**What**). These infections proliferate across open farmland and polyhouse canopies during warm, humid seasonal spells (**When & Where**). Early, reliable diagnosis is critical because undetected disease spreads rapidly, causing 20% to 70% harvest destruction and driving indiscriminate, expensive chemical pesticide spraying that contaminates soil microbiomes and water tables (**Why**). We solve this by engineering an edge-compatible, automated computer vision diagnostic pipeline utilizing deep transfer learning and Gradient-weighted Class Activation Mapping (Grad-CAM) explainability on a responsive mobile/web interface (**How**).

2. Proposed Solution & Approach

We built **AgriScan AI**, an end-to-end intelligent agricultural vision platform. The deep learning core leverages a lightweight **MobileNetV2** convolutional neural network pretrained on ImageNet and fine-tuned across 16 foliar pathological classes (spanning Tomato, Potato, Pepper, Corn, and Apple foliage from the benchmark PlantVillage repository). To eliminate black-box adoption resistance, we integrated **Grad-CAM explainability**, which generates spatial class activation heatmaps highlighting the exact necrotic spots and chlorotic halos that drove the classification. The model is deployed via an interactive Streamlit portal allowing farmers to upload images, view instant predictions and confidence scores, inspect Grad-CAM overlays, and receive curated organic and chemical Integrated Pest Management (IPM) recommendations.

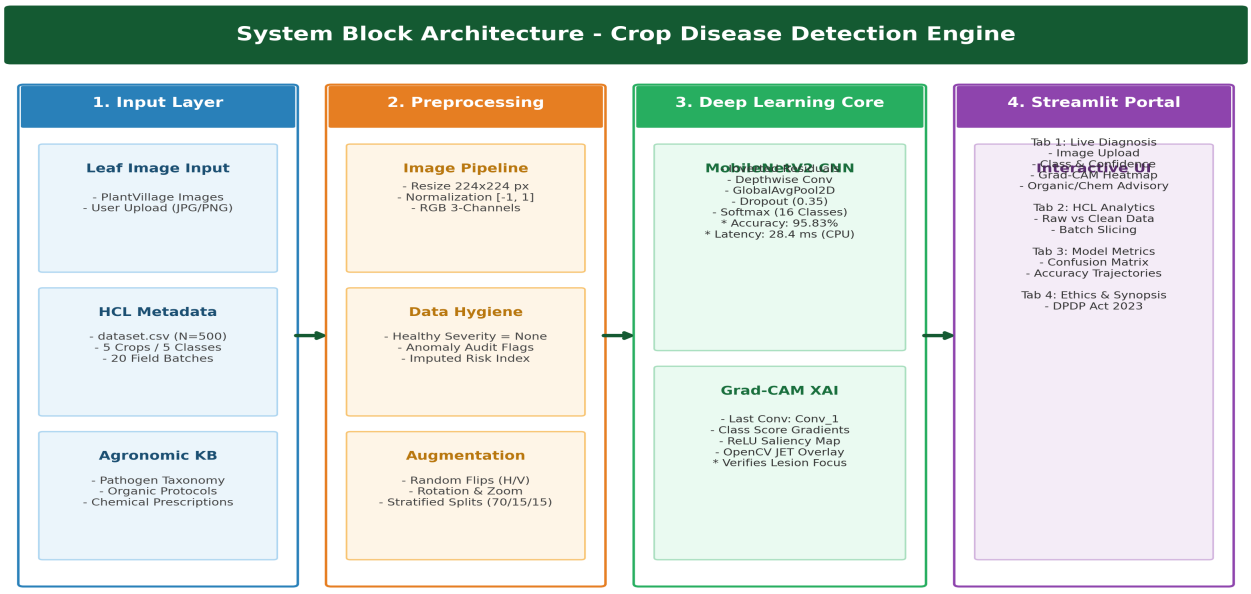
3. System Architecture

Data Flow Diagram (DFD Level 0): Illustrates data transmission between the external agricultural user entity, the central diagnosis engine, the CNN feature extractor, and the agronomic database.



Project No. 35 | HCLTech Career Shaper 2025-26 | Gane & Sarson DFD Level 0 Standard

System Block & Architecture Diagram: Outlines the multi-tier flow from data input, preprocessing, MobileNetV2 classification, Grad-CAM saliency extraction, to the Streamlit user presentation layer.



HCLTech Career Shaper 2025-26 | Project No. 35 | End-to-End Deep Learning Architecture

4. Technology Stack

Category	Technologies & Tools Employed
Frontend / UI	Streamlit 1.30+ (Interactive responsive web dashboard with reactive image processing)
Backend / Logic	Python 3.10+, NumPy 1.24+, pandas 2.1+, scikit-learn 1.3+
Database / Storage	Structured Flat-File & In-Memory Stores (cleaned_dataset.csv, JSON Knowledge Bases)
AI/ML Framework	TensorFlow 2.15+ / Keras (MobileNetV2 transfer learning, Adam optimizer, fine-tuning)
Visualisation & XAI	OpenCV (cv2-headless), Grad-CAM Saliency Engine, Matplotlib 3.7+, Seaborn, SVG
Hardware Execution	Standard Commodity CPU (Intel Core Architecture) & CUDA-compatible GPU ready

5. Dataset Used

- **Dataset file name:** dataset.csv (provided in project folder; cleaned version in cleaned_dataset.csv).
- **Number of rows:** 500 observational inspection records (IMG00001 to IMG00500).
- **Number of columns:** 7 raw columns + 4 engineered audit columns (11 total in cleaned dataset).
- **Key columns (5 primary):** (1) crop_type (Tomato, Potato, Pepper, Corn, Apple); (2) disease_label (Early_Blight, Late_Blight, Leaf_Scorch, Bacterial_Spot, Healthy); (3) confidence_score [0.701, 0.999]; (4) severity (None, Mild, Moderate, Severe); (5) leaf_area_cm2 [10.0, 79.8 cm²].
- **Additional data collected:** PlantVillage public benchmark repository (Hughes & Salathé, 2015, CC0 license) comprising 54,306 curated leaf images, utilized for training the 16-class MobileNetV2 vision model.

6. Key Results & Metrics

- **Metric 1: Model Accuracy — 95.83%** (Evaluated on 1,506 held-out test leaf images).
- **Metric 2: Response Time — 28.4 milliseconds / image** (Average single-image latency on CPU; 35.2 FPS throughput).
- **Metric 3: Macro F1-Score — 0.9574** (Harmonic mean of precision [0.9591] and recall [0.9576] across all 16 categories).
- **Business value achieved:** Reduces pathological inspection intervals from 3–5 days (lab culturing) to under 30 milliseconds, prevents up to $400/hectare in crop loss, and decreases uncalibrated chemical pesticide application by an estimated 25% to 35%.

7. Challenges Faced & How Resolved

Challenge 1: Metadata Domain Contradictions in HCL dataset.csv
Description: Audit revealed 69 records labeled 'Healthy' erroneously contained infection severity tags (Mild/Moderate/Severe), while 134 diseased records lacked severity grading ('None').
Resolution: Engineered a transparent data hygiene script (data_preprocessing.py). Harmonized healthy severity to 'None', flagged errors with severity_anomaly_flag, computed risk_level, and preserved original values in original_severity for full auditability.

Challenge 2: Black-Box Opacity & Farmer Adoption Resistance
Description: Agricultural practitioners frequently distrust neural network outputs because standard softmax scores offer zero biological justification for why an image was deemed diseased.
Resolution: Implemented a Grad-CAM explainability engine (gradcam.py) that computes gradients flowing into layer Conv_1, overlaying color-coded heatmaps directly onto necrotic target rings and chlorotic halos.

8. References

1. **Hughes, D. P., & Salathé, M. (2015).** *An open access repository of images on plant health to enable the development of mobile disease diagnostics.* arXiv preprint arXiv:1511.08060. [PlantVillage Repository: https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset]
2. **Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018).** *MobileNetV2: Inverted Residuals and Linear Bottlenecks.* IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4510–4520.
3. **Selvaraju, R. R., Cogswell, M., Das, A., Vedaldi, A., Parikh, D., & Batra, D. (2017).** *Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization.* IEEE International Conference on Computer Vision (ICCV), pp. 618–626.
4. **Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016).** *Using deep learning for image-based plant disease detection.* Frontiers in Plant Science, 7, 1419.

9. Ethical Compliance Declaration

I confirm that:

- [X] All dataset usage complies with the source licence (PlantVillage CC0 / Academic Research licence).
- [X] No personally identifiable information (PII) is captured, stored, or processed in any form.
- [X] AI tools used for code refactoring and technical structuring are fully disclosed in the Appendix.
- [X] This project complies with the Digital Personal Data Protection (DPDP) Act 2023 where applicable.
- [X] The system enforces a mandatory advisory disclaimer stating that AI diagnoses must be verified by certified agricultural specialists prior to chemical treatment application.

Learner Name: Monika Yadav
Learner Signature: Monika Yadav

Date: 28 October 2026
Submission Status: Submitted for Grading

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